

so-called gambler's ruin. The way to minimize that risk is to underbet, by using what is known as a *half-Kelly* or *fractional-Kelly* model. For example, if the Kelly model would tell you to bet 10 percent of your capital, you might choose to bet only 5 percent (half-Kelly) instead. Both the half-Kelly bet and the fractional-Kelly bet provide a margin of safety in portfolio management; that, together with the margin of safety we apply to selecting individual stocks, provides a double layer of protection.

Because the risk of overbetting far outweighs the penalties of underbetting, I believe that all investors, especially those who are just beginning to use a focus investment strategy, should use fractional-Kelly bets. Unfortunately, minimizing your bets also minimizes your potential gain. However, because the relationship in the Kelly model is parabolic, the penalty for underbetting is not severe. A half-Kelly bet, which reduces the amount you bet by 50 percent, reduces the potential growth rate by 25 percent.

Munger on Betting Odds

In 1994, Charlie Munger accepted an invitation from Dr. Guilford Babcock to address the student investment seminar at the University of Southern California's business school. He touched on several topics, including the "art of achieving worldly wisdom." He also explained, as only he can, what he thinks about probabilities and optimization.

"The model I like—to sort of simplify the notion of what goes on in a market for common stocks—is the pari-mutuel system at the racetrack," he said. "If you stop and think about it, a pari-mutuel system is a market. Everybody goes there and bets and the odds are changed based on what's bet. That's what happens in the stock market."

He continued, "Any damn fool can see that a horse carrying a light weight with a wonderful win rate and good position et cetera is way more likely to win than a horse with a terrible record and extra weight and so on and so on. But if you look at the odds, the bad horse pays 100 to 1, whereas the good horse pays 3 to 2. Then it's not clear which is statistically the best bet.